The 40% GC request — evidence, and what to do about it

Bottom line. 40% whole-fragment GC is arithmetically reachable but not advisable, and it does not solve the problem it is probably meant to solve. The GC-rich stretches in these constructs are concentrated in the **mEGFP reporter**, not in the therapeutic insert. A targeted recoding of the reporter removes 86% of them while keeping codon quality; a blanket 40% mandate removes them too, but degrades codon adaptation across every element and introduces sequence liabilities that were not there before.

1. Is 40% achievable?

Yes, arithmetically. The absolute synonymous floor for the engineered proprotein is **30.9% GC** — the value obtained by choosing the most AT-rich codon at every position. 40% sits above that floor, so no law of the genetic code forbids it.

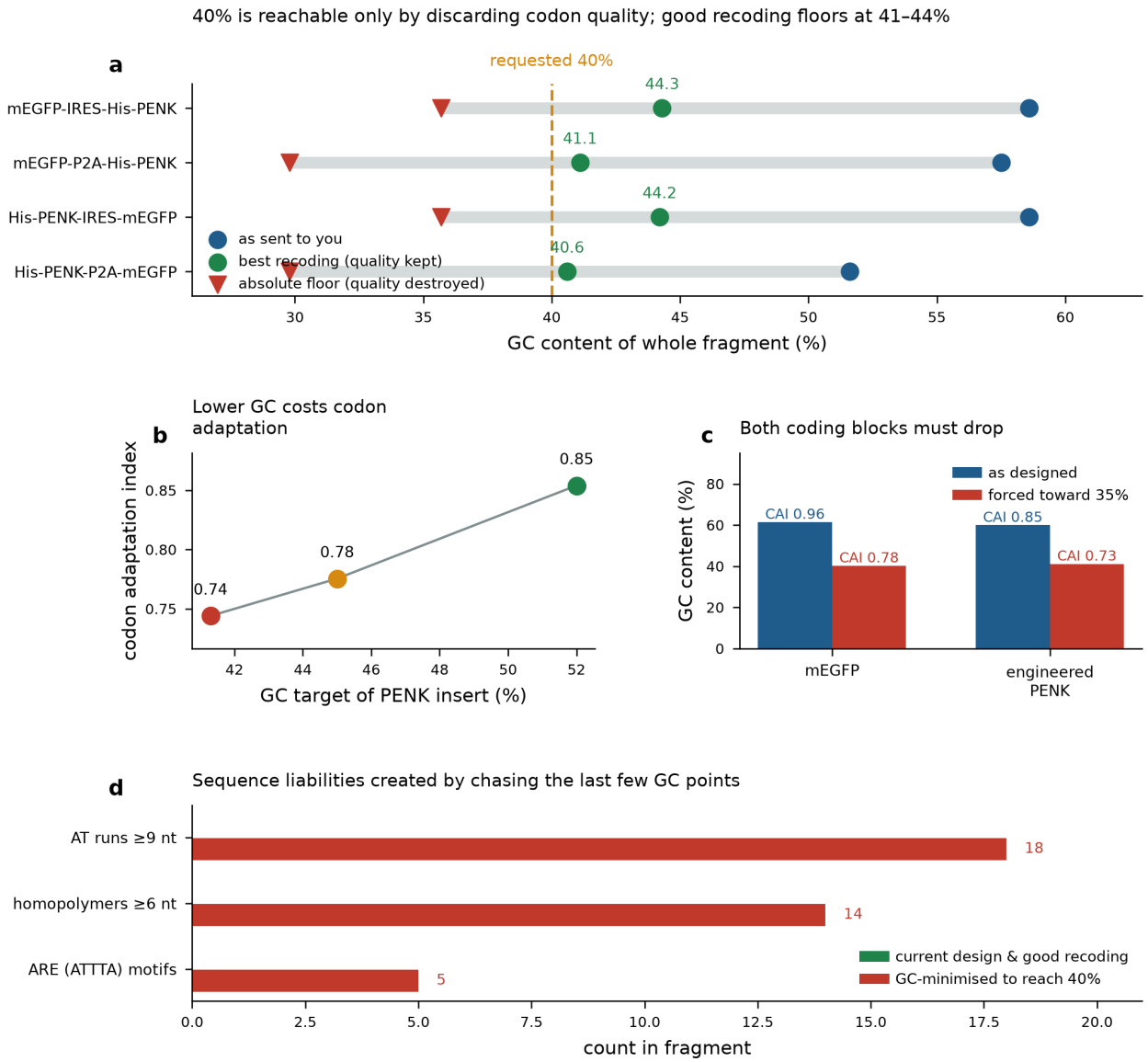
But the fragments contain a fixed element. The **IRES is not recodable** — it is a structured RNA element whose function depends on its actual sequence, not on the protein it encodes (it encodes nothing). At 53% GC across 574–584 nt, it drags the required GC of everything else downward:

fragment	total nt	fixed (IRES) nt	recodable nt	GC the recodable part must reach for 40% overall
mEGFP-IRES-His-PENK	2197	584	1613	35.2%
mEGFP-P2A-His-PENK	1674	10	1664	39.8%
His-PENK-IRES-mEGFP	2194	583	1611	35.2%
His-PENK-P2A-mEGFP	1671	9	1662	39.9%

And when the recoding is done **while preserving codon quality**, the fragments floor out above 40%:

fragment	GC as sent	best recoding with quality kept	GC-minimised floor (quality discarded)
mEGFP-IRES-His-PENK	58.6%	**44.3%**	35.7%
mEGFP-P2A-His-PENK	57.5%	**41.1%**	29.8%
His-PENK-IRES-mEGFP	58.6%	**44.2%**	35.7%
His-PENK-P2A-mEGFP	51.6%	**40.6%**	29.8%

So: the two IRES constructs **cannot** reach 40% without abandoning codon quality. The two P2A constructs can just about touch it.



2. What 40% costs

Codon adaptation. Codon adaptation index (CAI) measures how closely a coding sequence matches the synonymous codon preferences of the host [R3]. Driving the PENK insert down in GC drops CAI from **0.854 → 0.775 → 0.744** at 52%, 45% and 41% respectively. mEGFP is hit harder: **0.959 → 0.769**. Codon usage is an established determinant of expression level in mammalian cells, acting substantially through effects on mRNA levels rather than translation rate alone [R5, R6], and codon optimality is a major determinant of mRNA stability [R4].

GC content itself. This is the reference that argues most directly against the request: in mammalian cells, synonymous recoding to **higher** GC content increased mRNA levels and protein expression, with GC-rich versions of the same protein expressing severalfold better than GC-poor ones [R1]. A deliberate move from ~58% to 40% runs against that result.

Sequence liabilities. Reaching the GC floor by brute force introduces, per fragment:

liability	current design & good recoding	GC-minimised to reach 40%
AT runs ≥9 nt	0	18
homopolymers ≥6 nt	0	14
ARE (ATTTA) motifs	0	5

ATTTA pentamers in AU-rich context are the recognition core of the AU-rich element decay pathway, which targets transcripts for rapid deadenylation and degradation [R8, R9]. Creating five of them in a therapeutic transcript to satisfy a GC number is a poor trade. Long homopolymers and AT runs are also among the sequence features that raise synthesis difficulty and error rates [R11].

The general caution. Codon optimization in human therapeutics has a documented history of unintended consequences — altered folding, changed mRNA structure, loss of regulatory elements — and is not a free operation [R2]. mRNA structure independently regulates protein output through functional half-life [R10].

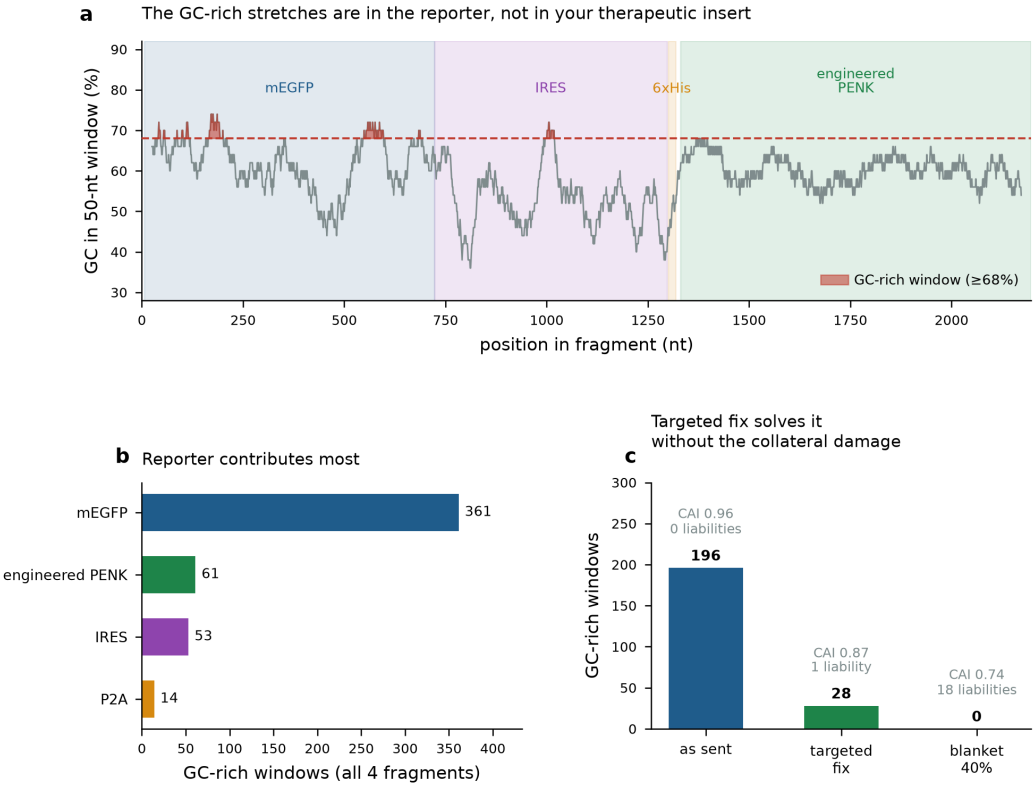
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3. The problem 40% is probably meant to solve

A vendor or synthesis flag on these fragments would most likely have been raised by **local GC-rich windows**, not by global GC. Counting 50-nt windows at ≥68% GC across all four fragments and attributing each to the element it falls in:

element	GC-rich windows (all 4 fragments)	recodable?
mEGFP	**361**	yes
engineered PENK	61	yes
IRES	53	**no — fixed sequence**
P2A	14	yes

The reporter accounts for 80% of the problem. Your own construct set already contains the natural experiment: `His-PENK-P2A-mEGFP` carries a lower-GC EGFP variant (41.0% GC) and has **zero** GC-rich windows in the reporter, versus 47 for the 53.5% mEGFP in `mEGFP-P2A-His-PENK`.



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4. Recommended alternative — targeted recoding

Recode the **reporter to 50% GC** and the **PENK insert to 52% GC**, leaving the IRES untouched:

fragment	GC as sent	after targeted fix	GC-rich windows before → after
mEGFP-IRES-His-PENK	58.6%	51.8%	196 → 28
mEGFP-P2A-His-PENK	57.5%	51.9%	75 → 14
His-PENK-IRES-mEGFP	58.6%	51.8%	203 → 25
His-PENK-P2A-mEGFP	51.6%	51.3%	15 → 0

The residual windows in the IRES constructs are **inside the IRES itself** and cannot be removed without destroying the element. Codon adaptation of the recoded mEGFP stays at 0.869 (from 0.959), against 0.769 if forced toward 35%. Cassette codon diversity is retained at 7/7 in the PENK insert.

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5. If the wet lab insists on 40% anyway

This is a judgement call that belongs to whoever owns the experiment, and there are defensible reasons to comply — a vendor quote conditioned on it, a synthesis platform with a hard specification, or simply not wanting to renegotiate. If the decision is to comply:

1. **Ask what the 40% is for, once.** If it came from a vendor rejection, ask for the actual flagged region. If the flag is a local GC-rich window, the targeted fix in §4 addresses it and the global number is beside the point. Most vendors accept 40–60% GC routinely, and these fragments at 51.6–58.6% are already inside that window.
2. **Apply it only where it is free.** The two P2A constructs reach 40.6–41.1% with codon quality largely intact. Do those, and exempt the IRES constructs, which cannot.
3. **Use the constrained builds, not a naive optimizer.** `engPENK_CDS_GC41_absolute_best_at_40pct` reaches 41.3% with CAI 0.744, zero homopolymers, zero AT runs ≥9 nt and zero ARE motifs. A most-AT-codon pass reaching 29.8% produces 18 AT runs, 14 homopolymers and 5 AREs. The difference between these two is entirely in how the optimizer is constrained.
4. **Do not touch the signal peptide region or the cassettes.** Verified: all four candidate sequences translate residue-for-residue to the intended proprotein, all 7 YGGFEVVG cassettes retain distinct codon usage, and the signal peptide is unaltered.
5. **Treat expression as an empirical question.** None of this predicts protein yield. If the group wants to know whether a 40% version expresses as well as a 52% version, that is a transfection and a western blot, not a calculation — and given [R1], it is a result worth having before committing the whole construct set.

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6. Separate and more urgent — the His tag placement

Independent of GC, the audit found a design issue in all four fragments that outranks the GC question: **the 6×His tag sits N-terminal to the signal peptide.** Residues upstream of the signal peptide start:

fragment	residues before signal peptide
mEGFP-IRES-His-PENK	12
mEGFP-P2A-His-PENK	**268**
His-PENK-IRES-mEGFP	11
His-PENK-P2A-mEGFP	11

A signal peptide is recognised co-translationally at the extreme N-terminus of the nascent chain; placing residues in front of it interferes with recognition and cleavage [R13]. For the P2A construct the displacement is the entire upstream reporter plus the 2A peptide — and a gene downstream of a 2A element requires its own signal sequence to be secreted, because 2A "self-cleavage" leaves the

downstream product with an N-terminal proline rather than a free signal peptide [R12]. Moving the tag to the C-terminus, or placing it immediately after the signal peptide cleavage site, resolves this. It should be settled before any resynthesis, since resynthesising at 40% GC would lock in the tag error.

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Computational analysis only. GC content, codon adaptation index, local window scans and motif counts are sequence-derived measures; they do not predict expression level, secretion efficiency or protein yield, all of which require measurement.